Explain basis of common GI disturbances

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Common gastrointestinal (GI) disturbances stem from disrupted motility, abnormal gut sensitivity, an unbalanced microbiome, or structural anomalies. The exact drivers vary depending on whether the condition is functional (a brain-gut interaction issue) or organic (a physical disease).

Frequent Conditions

Gastroesophageal Reflux Disease (GERD): Chronic acid reflux that irritates the lining of the esophagus.

Irritable Bowel Syndrome (IBS): A common functional disorder characterized by cramping, abdominal pain, bloating, gas, and alternating bouts of diarrhea or constipation.

Gastroenteritis: Often called the "stomach flu," this is an intestinal infection caused by a virus, bacteria, or parasites.

Celiac Disease: An autoimmune reaction to eating gluten that damages the small intestine.

Inflammatory Bowel Disease (IBD): Includes conditions like Crohn's disease and Ulcerative Colitis, which involve chronic, destructive inflammation of the digestive tract.

- Common Gastro Intestinal disturbances include
- Disorders of Swallowing and the Esophagus Paralysis of the Swallowing Mechanism
- Achalasia and Mega esophagus
- Disorders of the Stomach Gastritis Chronic Gastritis
- Achlorhydria (and Hypochlorhydria).
- Gastric Atrophy May Cause Pernicious Anemia
- Peptic Ulcer
- Disorders of the Small Intestine Abnormal Digestion of Food in the Small Intestine—
- Tropical Sprue.

Disorders of the Large Intestine Constipation Disorders of the Large Intestine Constipation

- Megacolon (Hirschsprung's Disease).
- Diarrhea
- Ulcerative Colitis.

Disorders of Swallowing and the Esophagus Paralysis of the Swallowing Mechanism

Disorders of Swallowing and of the Esophagus

Dysphagia or difficulty swallowing is a symptom of many different medical conditions. These conditions include nervous system and brain disorders, muscle disorders and physical blockages in your throat.

Paralysis of the Swallowing Mechanism.

Damage to the 5th, 9th, or 10th cerebral nerve can cause paralysis of significant portions of the swallowing mechanism.

Paralysis of the swallowing muscles, as occurs in persons with muscle dystrophy or as a result of failure of neuromuscular transmission in persons with myasthenia gravis or botulism, can also prevent normal swallowing.

Types of dysphagia

There are three types of the dysphagia

Oral dysphagia:

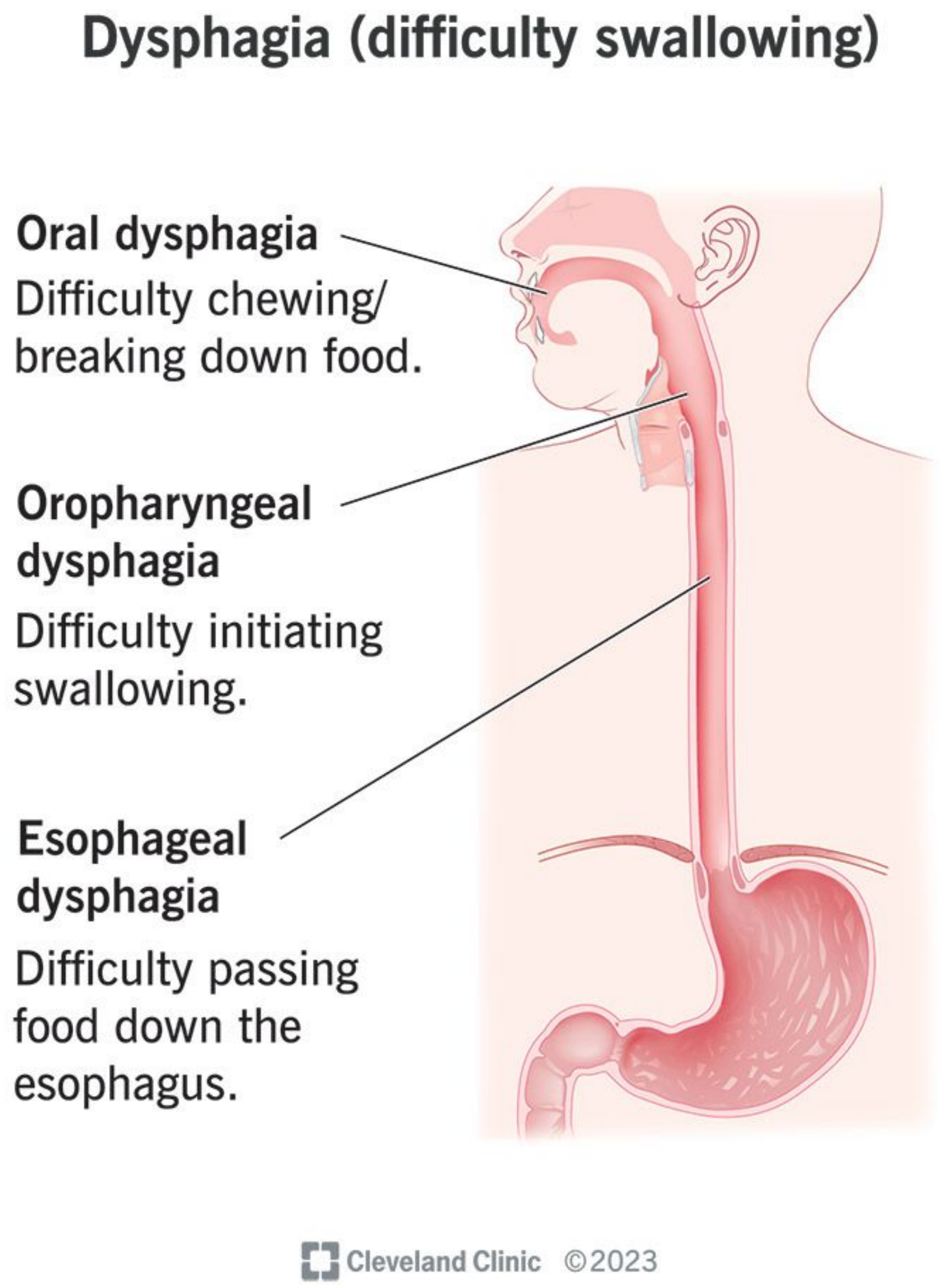
The problem is in mouth. The jaw, teeth and tongue work together to tear food into smaller pieces when you chew. Salivary glands produce spit that softens the food so it breaks apart easily.

Oropharyngeal dysphagia:

The problem is in throat. After mouth prepares the food, tongue pushes it to the back of throat. voice box (larynx) closes to prevent food or liquid from slipping into airway (trachea) on its way to esophagus.

Esophageal dysphagia:

The problem is in esophagus. Esophagus squeezes the food or liquid down in a wave-like motion (peristalsis) until it reaches your stomach.




DIFFICULTY SWALLOWING (DYSPHAGIA)
CAUSES OF ESOPHAGEAL DYSPHAGIA
GERD
ACHALASIA
DIFFUSE ESOPHAGEAL SPASM
ESOPHAGEAL CANCER
SCLERODERMA
ESOPHAGEAL STRICTURE
ESOPHAGEAL DYSPHAGIA

MelbourneENT

SYMPTOMS OF DYSPHAGIA

- Drooling
- Hoarseness
- Regurgitation (bringing food back up)
- Choking while eating
- Unexplained weight loss
- Difficulty controlling food in the mouth
- Recurring heartburn
- Recurring pneumonia
- Not able to control saliva in the mouth
- Problem with starting the swallowing process
- Gagging or coughing while swallowing
- Acid from the stomach backing up into the throat
- Feeling as if food is stuck in chest, throat, behind the sternum, or breastbone

Symptoms of Dysphagia
Pain or discomfort while swallowing
Feeling that food is stuck in your chest or throat
Coughing or choking while eating or drinking
Frequent drooling
Regurgitation (bringing food back up)
Hoarseness or voice changes
Unexplained weight loss
Heartburn or indigestion
Nasal regurgitation (food coming out through the nose)

Achalasia and Mega esophagus.

Achalasia is a rare esophageal motility disorder. Achalasia is a condition in which the lower esophageal sphincter fails to relax during swallowing. As a result, food swallowed into the esophagus fails to pass from the esophagus into the stomach.

Damaged nerves make it hard for the muscles of the esophagus to squeeze food and liquid into the stomach.
Achalasia: Characterized by the absence of normal esophageal contractions (peristalsis) and the inability of the lower esophageal sphincter (LES) to open during swallowing.

Mega esophagus is a rare complication of achalasia characterized by severe dilatation of the esophagus, often indicative of end-stage achalasia.

When achalasia becomes severe, the esophagus often cannot empty the swallowed food into the stomach for many hours, instead of the few seconds that is the normal time. Over months and years, the esophagus becomes tremendously enlarged until it often can hold as much as 1 liter of food, which often becomes putridly infected during the long periods of esophageal stasis.

Gastritis

Disorders of the Stomach

Gastritis—Gastritis is an inflammation of the stomach lining. The stomach lining is a mucus-lined barrier that protects the stomach wall. Weaknesses or injury to the barrier allows digestive juices to damage and inflame the stomach lining. Several diseases and conditions can increase the risk of gastritis.

Inflammation of the Gastric Mucosa Mild to moderate chronic gastritis is especially common in the middle to later years of adult life.

The inflammation of gastritis may be only superficial and therefore not very harmful, or it can penetrate deeply into the gastric mucosa, in many long- standing cases causing almost complete atrophy of the gastric mucosa.

In a few cases, gastritis can be acute and severe, with ulcerative excoriation of the stomach mucosa by the stomach's own peptic secretions.

Gastritis is an inflammation, irritation, or erosion of the protective stomach lining. It can be acute (developing suddenly) or chronic (developing slowly over time). Common symptoms include burning

Common Symptoms

While some cases are asymptomatic, those affected frequently experience:

upper abdominal pain, bloating, and indigestion.

A burning, gnawing ache or pain in the upper abdomen (which may worsen or improve after eating)

Nausea or recurrent vomiting

Indigestion and a feeling of fullness in the upper abdomen, especially after a small meal

Loss of appetite

Causes

The stomach's mucus lining protects it from harsh digestive juices. Gastritis occurs when this barrier weakens, allowing acid to damage the stomach wall. Major causes and risk factor include:

Factors that increase your risk of gastritis include:

Bacterial infection.

A bacterial infection called Helicobacter pylori, also known as H. pylori, is one of the most common worldwide human infections.

Regular use of pain relievers.

Pain relievers known as nonsteroidal anti-inflammatory drugs, also called NSAIDs, can cause both acute gastritis and chronic gastritis.

Older age.

Older adults have an increased risk of gastritis because the stomach lining tends to thin with age. Older adults also have an increased risk because they are more likely to have H. pylori infection or autoimmune disorders than younger people are.

Excessive alcohol use.

Alcohol can irritate and break down your stomach lining. This makes your stomach more vulnerable to digestive juices. Excessive alcohol use is more likely to cause acute gastritis.

Stress.

Severe stress due to major surgery, injury, burns or severe infections can cause acute gastritis.

Cancer treatment.

Chemotherapy medicines or radiation treatment can increase your risk of gastritis.

Autoimmune gastritis

It is more common in people with other autoimmune disorders. These include Hashimoto's disease and type 1 diabetes. Autoimmune gastritis also can be associated with vitamin B-12 deficiency.

Other diseases and conditions.

Gastritis may be associated with other medical conditions. These may include HIV/AIDS, Crohn's disease, celiac disease, sarcoidosis and parasitic infections

What are the different types of gastritis?

There are to type ,acute gastritis or chronic gastritis.

Acute gastritis is sudden and temporary. The conditions that cause it are also acute.

Chronic gastritis is a long-term condition, though you may not notice it all the time (or at all). It tends to develop gradually, as a result of another chronic condition.

Gastritis can also be erosive or non-erosive.

Erosive gastritis means the thing that’s causing your gastritis is actually eating away at your stomach lining, leaving wounds (ulcers). It’s often a chemical, like acid, bile, alcohol or drugs.

Nonerosive gastritis doesn’t leave erosive changes but may cause irritation, such as reddening of the stomach lining.

Gastritis may go by a more specific name, based on the cause. Some examples are:

- Infectious gastritis.
- Drug-induced gastritis.
- Alcohol-induced gastritis.
- Stress-induced gastritis.
- Autoimmune gastritis.
- Eosinophilic Gastritis

Management & Treatment

Treatment primarily focuses on reducing stomach acid, fighting infections, and eliminating irritants:

Medications: Doctors often prescribe antacids, H2 blockers, or proton pump inhibitors (PPIs) like Nexium to neutralize or reduce acid. If *H. pylori* is present, a course of antibiotics will be required.

Dietary Adjustments: Avoiding irritants such as alcohol, caffeine, spicy foods, and high-fat items can alleviate flare-ups.

Complication

Chronic Gastritis Can Lead to Gastric Atrophy and Loss of Stomach Secretions

Achlorhydria (and Hypochlorhydria). Achlorhydria means that the stomach fails to secrete hydrochloric acid;

Loss of the stomach secretions in gastric atrophy leads to achlorhydria and, occasionally, to pernicious anemia. **Pernicious Anemia.**

Gastric Atrophy May Cause Pernicious anemia commonly accompanies gastric atrophy and achlorhydria.

Peptic ulcers

Stomach cancer

Chronic H. pylori gastritis and autoimmune gastritis increase the chance of developing growths in the stomach lining. These growths may be benign or may be stomach cancer

Peptic Ulcer

A peptic ulcer is an excoriated area of stomach or intestinal mucosa caused principally by the digestive action of gastric juice or upper small intestinal secretions.1 shows the points in the gastrointestinal tract at which peptic ulcers most frequently occur, demonstrating that the most frequent site is within a few centimeters of the pylorus. In addition, peptic ulcers frequently occur along the lesser curvature of the antral end of the stomach or, more rarely, in the lower end of the esophagus where stomach

Causes:

1. High acid and pepsin content
2. Irritation
3. Poor blood supply
4. Poor secretion of mucus
5. Infection (H. pylori)

Therefore, a peptic ulcer can be caused in either of two ways: (1) excess secretion of acid and pepsin by the gastric

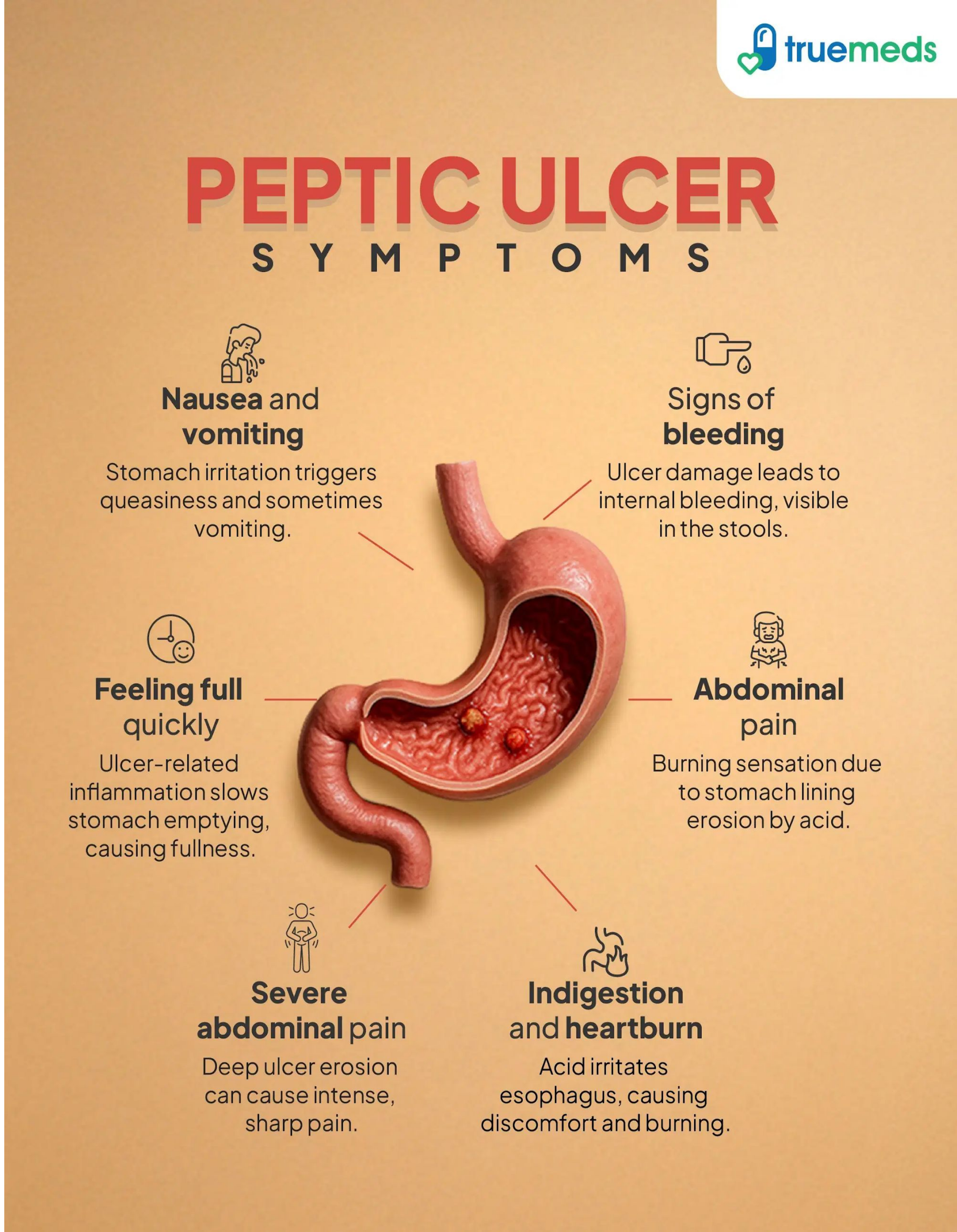
mucosa or (2) diminished ability of the gastroduodenal mu cosal barrier to protect against the digestive properties of the stomach acid–pepsin secretion.

Symptoms

- Signs and symptoms can vary, but the most common include:
- A burning or gnawing sensation in the stomach area (pain is often worse between meals or at night, and may briefly ease after eating or taking antacids).
- Bloating and a feeling of fullness.
- Nausea or vomiting.
- Unexplained weight loss.

Diagnosis

- If you suspect you have an ulcer, a gastroenterologist may use one of the following diagnostic tools:
- **Upper Endoscopy:** A procedure where a small camera is inserted down your throat to visually inspect the lining of your stomach and intestine.
- **H. pylori Tests:** Blood, breath, or stool tests to check for bacterial infection.



Treatment

Treatment focuses on eliminating the cause, relieving symptoms, and healing the sore. Common treatment methods include:

Antibiotics: Required if an *H. pylori* infection is found use of antibiotics along with other agents to kill infectious bacteria;

Proton Pump Inhibitors (PPIs):

Administration of an acid- suppressant drug, especially ranitidine, which is an antihistaminic agent that blocks the stimulatory effect of histamine on gastric gland histamine2 receptors, thus reducing gastric acid secretion by 70% to 80%. These physiological approaches to therapy have proven to be effective in most patients.

Lifestyle Changes: Quitting smoking, limiting alcohol, and switching pain medications if NSAIDs are the root cause.

In a few cases, however, the patient’s condition is so severe, including massive bleeding from the ulcer, that heroic operative procedures must be used;

these procedures include removal of part of the stomach or cutting the two vagus nerves that supply parasympathetic stimulation to the gastric glands.

Complications

Untreated peptic ulcers can cause:

Bleeding in the stomach or duodenum.

Bleeding can be a slow blood loss that leads to too few red blood cells, called anemia. Or you can lose enough blood so that you need to be in a hospital or get blood from a donor. Severe blood loss may cause black or bloody vomit or black or bloody stools.

A hole, called a perforation, in the stomach wall.

Peptic ulcers can eat a hole through the wall of your stomach or small intestine. This puts you at risk of infection of your abdomen, called peritonitis.

Blockage. Peptic ulcers can keep food from going through the digestive tract. The blockage can make you feel full easily and cause you to vomit and lose weight.

Stomach cancer. Studies have shown that people infected with H. pylori have an increased risk of stomach cancer.

Prevention

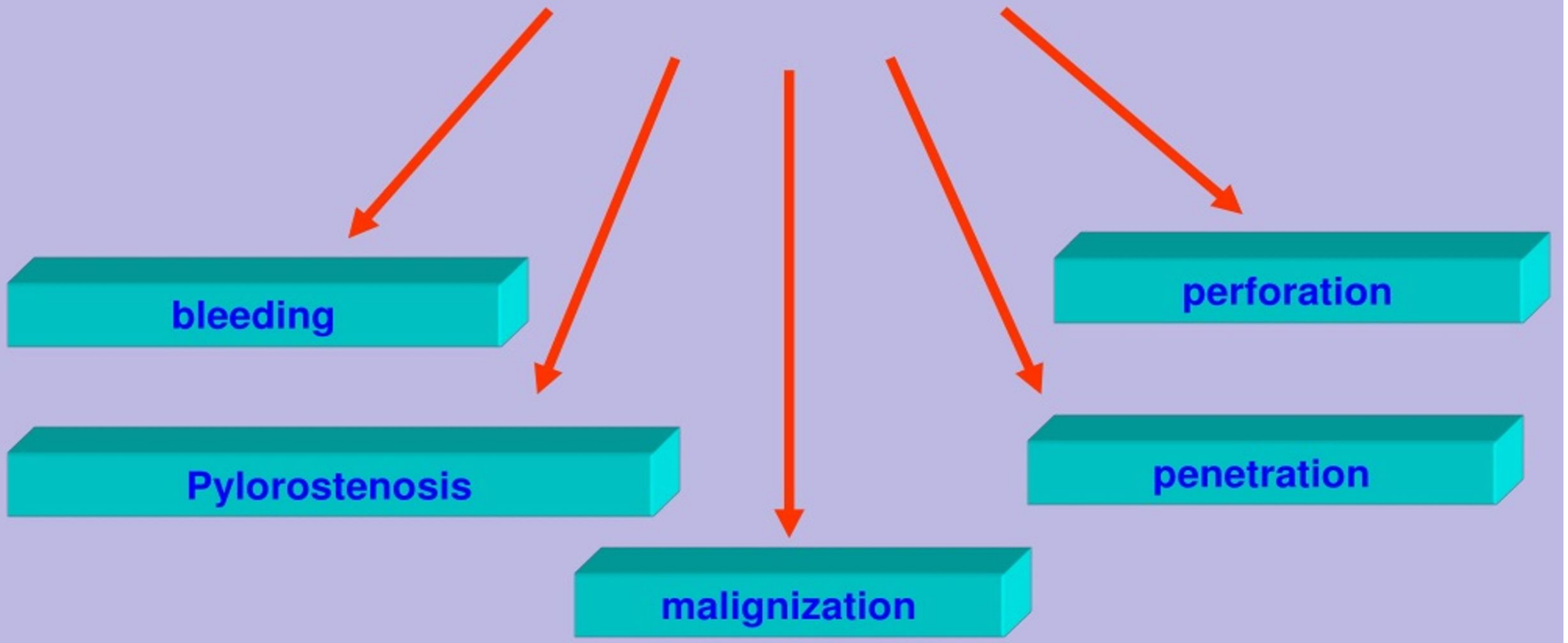
To help prevent peptic ulcers:

Take care with pain relievers. If you often use NSAIDs, which can increase your risk of peptic ulcer, take steps to reduce your risk of stomach problems. For instance, take pain relievers with meals.

If you need an NSAID, you also may need to take other medicines to help protect your stomach. These include antacids, proton pump inhibitors, acid blockers or cytoprotective agents.

If you smoke, find a way to quit. Quitting smoking can lower your risk of peptic ulcer. Talk with your healthcare professional for help with quitting.

Complications of the ulcer disease



Disorders of the Small Intestine

Abnormal Digestion of Food in the Small Intestine—

It may be due to

A-Pancreatic Failure:-

A serious cause of abnormal digestion is failure of the pancreas to secrete pancreatic juice into the small intestine. Lack of pancreatic secretion frequently occurs

(1) in persons with pancreatitis ,

(2) when the pancreatic duct is blocked by a gallstone at the papilla of Vater, or

(3) after the head of the pancreas has been removed because of malignancy.

Loss of pancreatic juice means loss of trypsin, chymotrypsin, carboxypolypeptidase, pancreatic amylase, pancreatic lipase, and a few other digestive enzymes. Without these enzymes, up to 60% of the fat entering the small intestine may not be absorbed, along with one-third to one-half of the proteins and carbohydrates. As a result, large portions of the ingested food cannot be used for nutrition and copious, fatty feces are excreted.

B-Pancreatitis—Inflammation of the Pancreas.

Pancreatitis can occur in the form of either acute pancreatitis or chronic pancreatitis.

The most common cause of pancreatitis is drinking excess alcohol, and the second most common cause is blockage of the papilla of Vater by a gallstone.

Other factors than can con tribute to chronic pancreatitis include smoking, high levels of triglycerides, or autoimmune- mediate inflammation.

Malabsorption by the Small Intestinal Mucosa

Sprue :-

Occasionally, nutrients are not adequately absorbed from the small intestine even though the food has been well digested. Several diseases can cause decreased absorption by the mucosa; they are often classified together under the general term "sprue."

Malabsorption also can occur when large portions of the small intestine have been removed.

There are two types of sprue.

Non tropical Sprue OR (celiac disease)

Celiac disease (also called non-tropical sprue) is an intestinal disorder in which the body cannot tolerate gluten. Gluten is a natural protein in many grains, including wheat, barley, rye, and oats. People with celiac disease have an immune reaction that is triggered by gluten. The immune reaction causes inflammation at the surface of the small intestine where it damages small structures (villi) on the surface of the intestine. It also damages smaller, hair-sized protrusions called microvilli. Healthy villi and microvilli are needed for normal digestion. When they are damaged, the intestine cannot absorb nutrients properly and you can become malnourished.

Tropical sprue

It is a malabsorption syndrome characterized by chronic diarrhea, nutrient deficiencies (particularly of folate and vitamin B12), and weight loss. It typically affects people who live in or travel to tropical and subtropical regions, with an infectious origin suspected

Malabsorption in Sprue.

In the early stages of sprue, intestinal absorption of fat is more impaired than is absorption of other digestive products.

The fat that appears in the stools is almost entirely in the form of salts of fatty acids rather than undigested fat, demonstrating that the problem is one of absorption, not of digestion. In fact, the condition is frequently called **steatorrhea,** which means simply excess fats in the stools.

In severe cases of sprue, in addition to malabsorption of fats, impaired absorption of proteins, carbohydrates, calcium, vitamin K, folic acid, and vitamin B12 also occurs.

As a result, the person experiences the following: (1) severe nutritional deficiency, which often results in wasting of the body;

(2) osteomalacia (i.e., demineralization of the bones because of lack of calcium);

(3) inadequate blood coagulation caused by lack of vitamin K;

(4) macrocytic anemia of the pernicious anemia type, resulting from diminished vitamin B12 and folic acid absorption.

Disorders of the Large Intestine

Constipation is typically defined as having fewer than three bowel movements a week, or passing stools that are hard, dry, lumpy, and difficult to pass.

Constipation is usually caused by a lack of dietary fiber, insufficient fluid intake, or physical inactivity, and can generally be managed at home.

Constipation means slow movement of feces through the large intestine.

Constipation is often associated with large quantities of dry, hard feces in the descending colon that accumulate because of excess absorption of fluid or insufficient fluid intake.

Any pathology of the intestines that obstructs movement of intestinal contents, such as tumors, adhesions that constrict the intestines, or ulcers, can cause constipation. Infants are seldom constipated, but part of their training in the early years of life requires that they learn to control defecation; this control is effected by inhibiting the natural defecation reflexes.

- **Lifestyle causes**
- Slow stool movement may happen when a person does not:
- Drink enough fluids.
- Eat enough dietary fiber.
- Exercise regularly.
- Use the toilet when there's an urge to pass stool.
- **Complications**
- Complications of chronic constipation include:
- Swollen tissues around the anus, also called hemorrhoids.
- Torn tissues of the anus, also called anal fissures.
- Hard stools backed up into the colon, also called fecal impaction.
- Exposed tissues of the rectum that have slipped out of the anal opening, also called rectal prolapse.

Megacolon (Hirschsprung's Disease).

Hirschsprung's disease (congenital aganglionic megacolon) is a condition present at birth where nerve cells are missing in the lower large intestine. This causes severe bowel blockage, as waste cannot be pushed through.

Occasionally, constipation is so severe that bowel movements occur only once every several days or sometimes only once a week.

This phenomenon allows tremendous quantities of fecal matter to accumulate in the colon, causing the colon sometimes to distend to a diameter of 3 to 4 inches. The condition is called mega colon, or Hirschsprung's disease

Diarrhea

Diarrhea is passing loose, watery stools three or more times a day. It is usually a short-term illness caused by viral or bacterial infections. The primary goal of managing diarrhea is to stay hydrated and replace lost electrolytes.

It results from rapid movement of fecal matter through the large intestine.

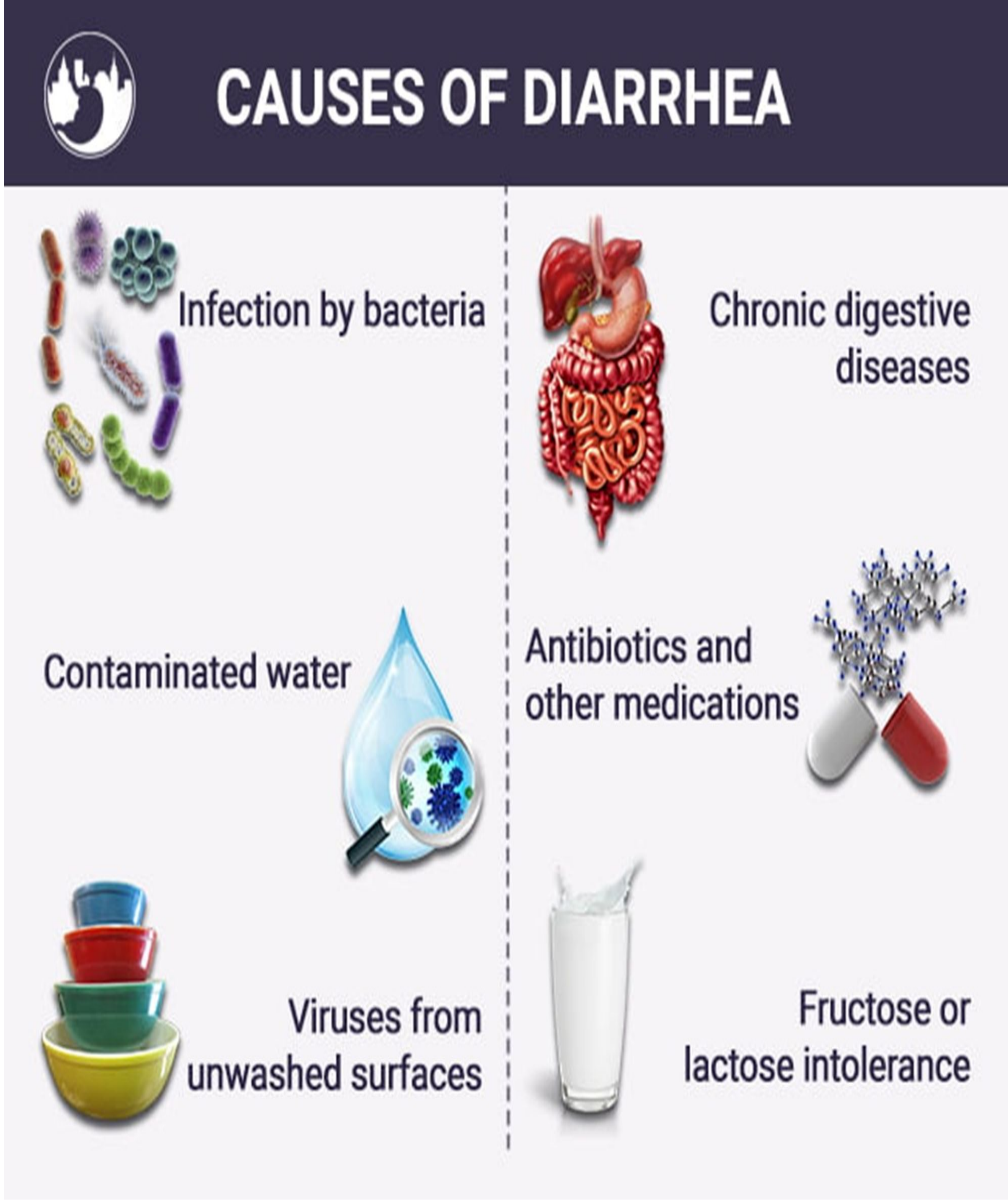
Causes

Enteritis:- Enteritis means inflammation of the Intestinal tract usually caused either by viruses or bacteria in the intestinal tract Infections: Viruses (like rotavirus or norovirus), bacteria (such as E. coli), and parasites are the most common culprits.

Cholera is an acute intestinal infection caused by the bacterium Vibrio cholera, which is transmitted primarily through contaminated food or water. It triggers severe, watery diarrhea and vomiting that can lead to life-threatening dehydration and electrolyte imbalance within hours

Medications: Certain drugs, especially antibiotics, can disrupt your gut flora.

Diet & Sensitivities: Food intolerances (like lactose intolerance), artificial sweeteners, or spicy foods can trigger loose stools.



Underlying Conditions: Chronic diarrhea (lasting longer than four weeks) can be a sign of conditions like Irritable Bowel Syndrome (IBS), Crohn's disease, or celiac disease

Therefore, loss of fluid and electrolytes can be so debilitating within several days that death can ensue.

The most important physiological basis of therapy in diarrhea (cholera) is to replace the fluid and electrolytes as rapidly as they are lost, mainly by giving the patient intravenous solutions. With proper therapy, along with the use of antibiotics, almost no persons with cholera die, but without therapy, up to 50% of patients die.

Psychogenic Diarrhea. Most people are familiar with the diarrhea that accompanies periods of nervous tension, such as during examination time or when a soldier is about to go into battle. This type of diarrhea, called psychogenic emotional diarrhea, is caused by excessive stimulation of the parasympathetic nervous system, which greatly excites both (1) motility and (2) excess secretion of mucus in the distal colon. These two effects added together can cause marked diarrhea.

What are the complications of diarrhea?

Dehydration is one of the biggest concerns with diarrhea.

Without treatment, dehydration can lead to

kidney failure,

stroke,

heart attack

or even death


HMD
RECOGNISING THE SIGNS
COMMON SYMPTOMS OF DIARRHOEA
Fever above 39°C
Pressing bowel urgency
Dehydration
Blood and mucus in stool
Vomiting
Constant abdominal pain
Bloated stomach

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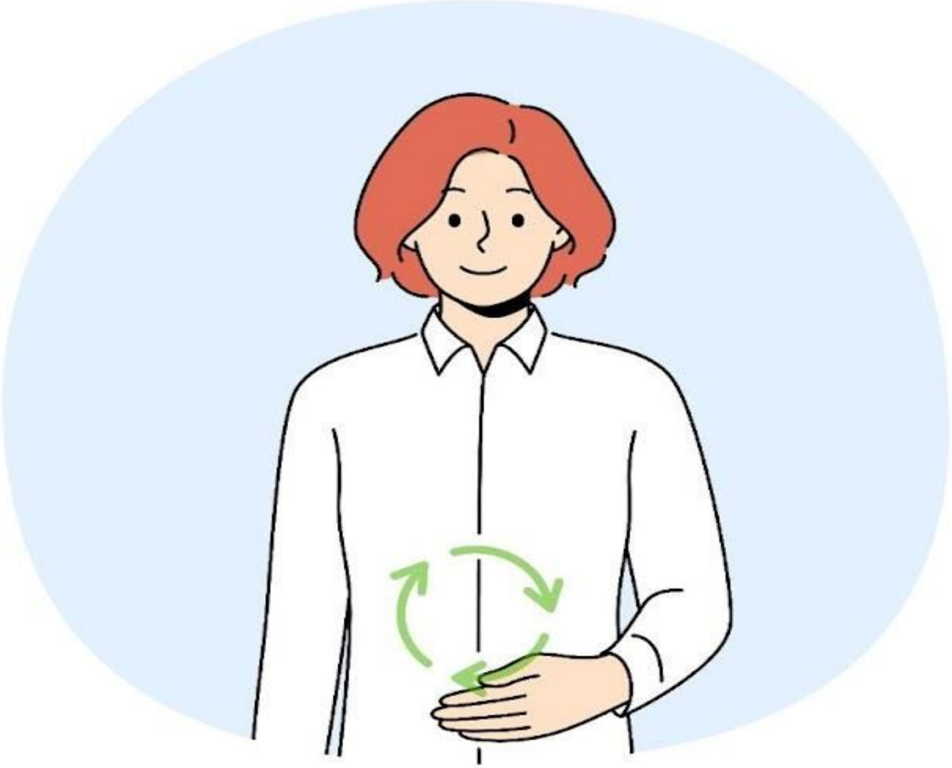
Symptoms of Diarrhea/Loose Motion



- Frequent Watery Stools
- Abdominal Cramps
- Urgent Need to Use the Toilet
- Bloating
- Nausea and Vomiting
- Dehydration
- Fever and Chills
- Weakness and Fatigue

Ulcerative colitis is a chronic inflammatory bowel disease (IBD) that causes long-lasting inflammation and ulcers (sores) in the innermost lining of the large intestine (the colon and rectum). It leads to painful flare-ups interspersed with periods of remission.

Causes & Risk Factors

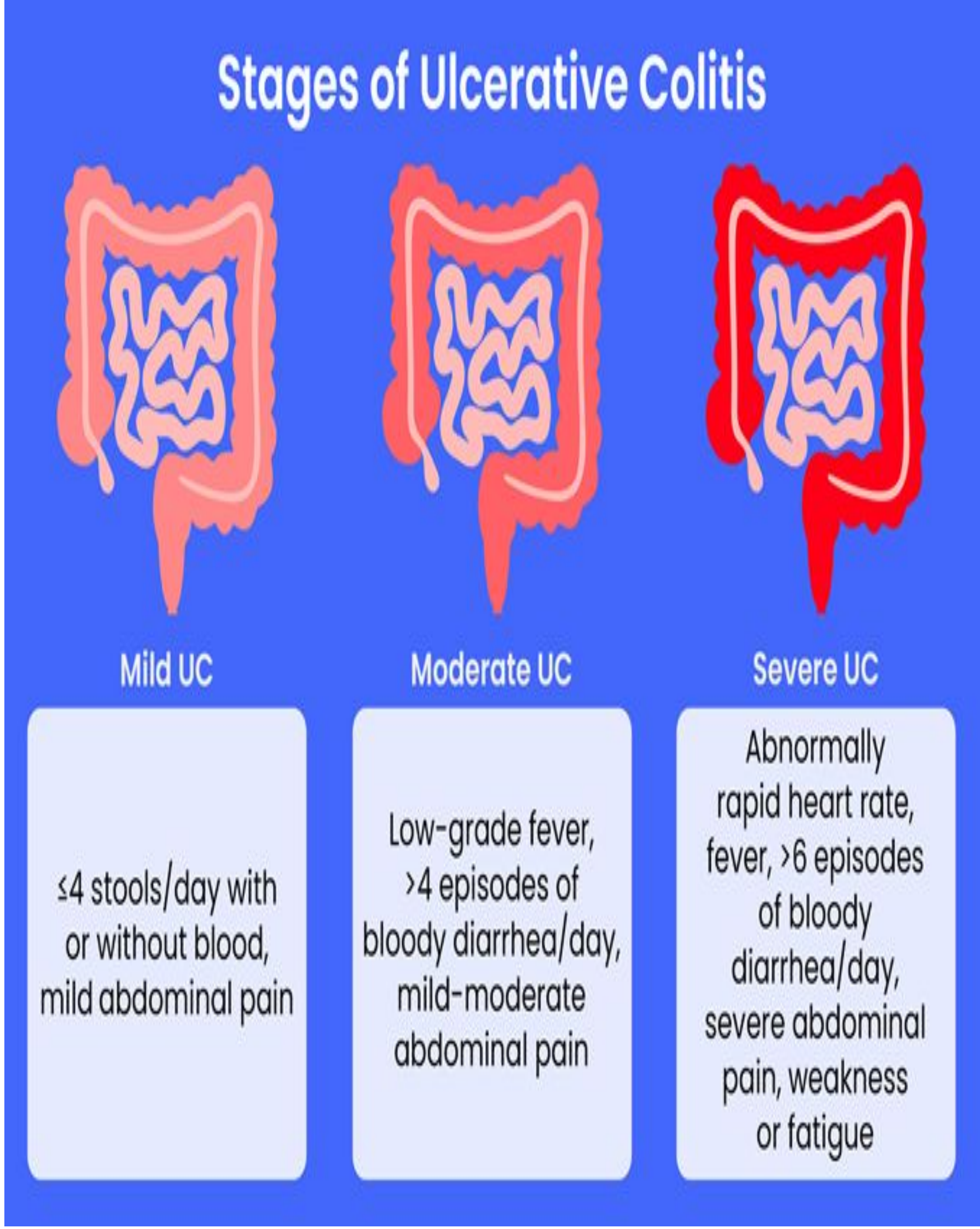
The exact cause is unknown, but medical experts point to a combination of factors:

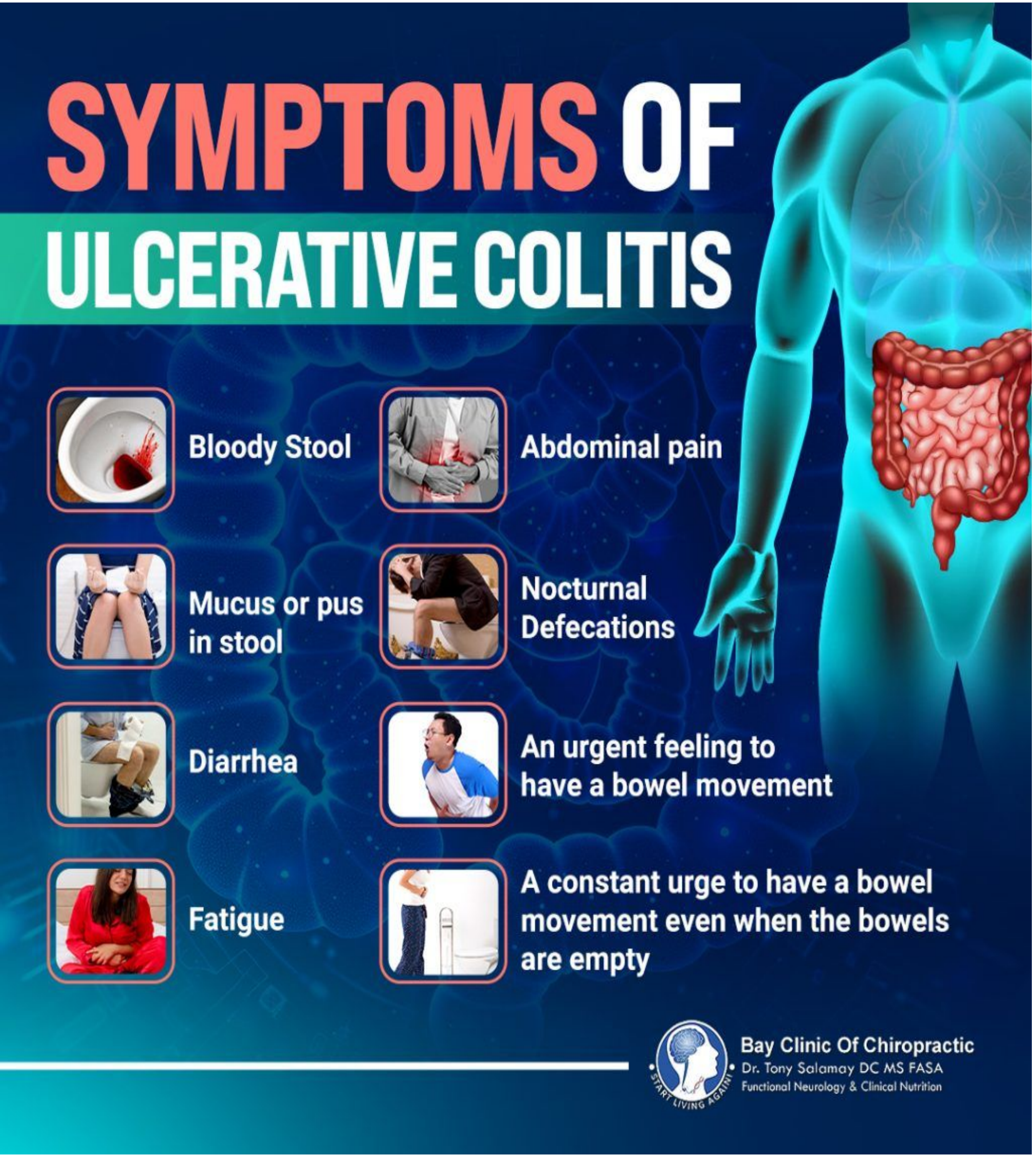
Immune System Malfunction: The immune system mistakenly attacks the digestive tract.

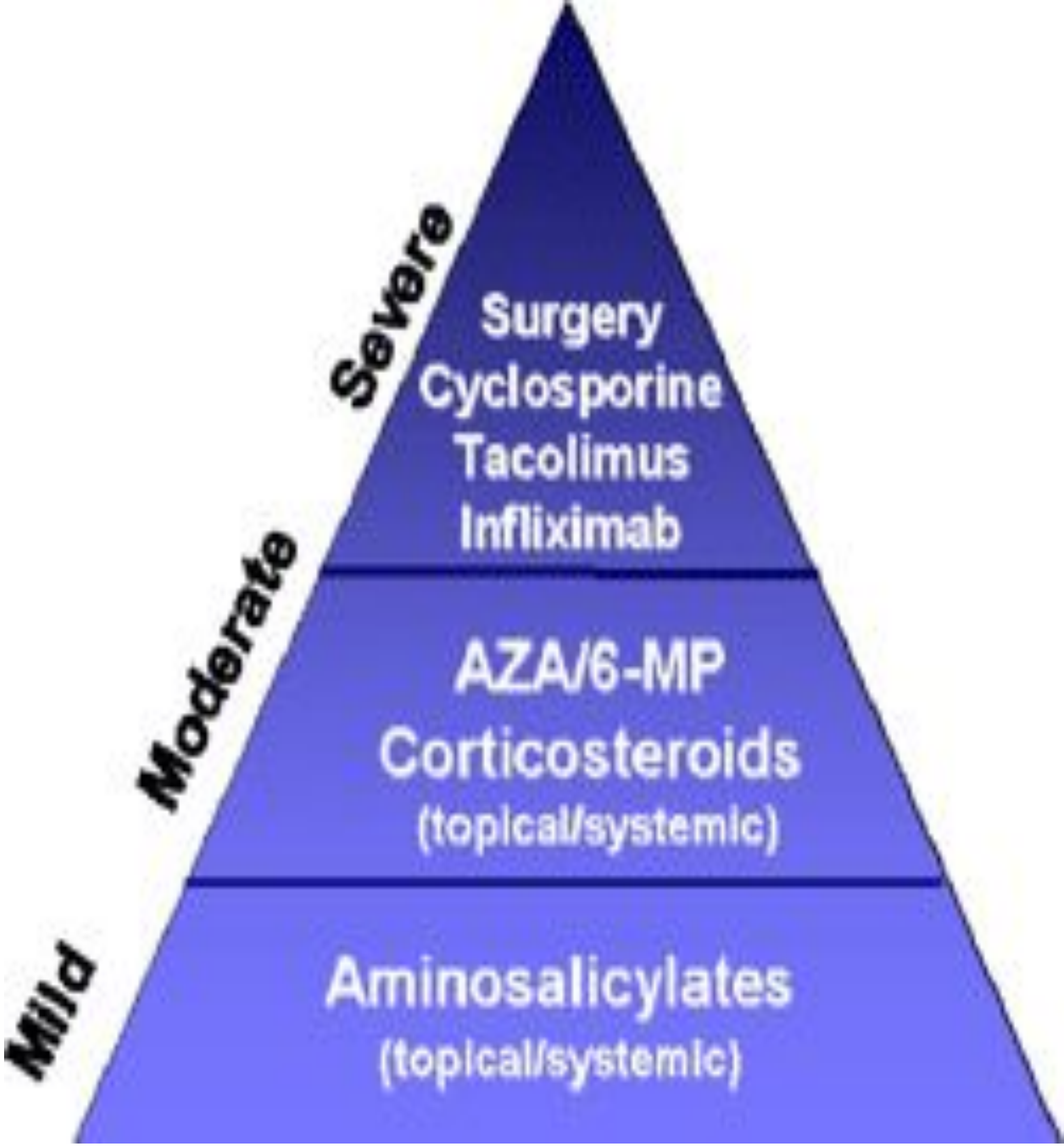
Genetics: It often runs in families.

Environmental Factors: Diet, stress, and microbiome imbalances may trigger the disease or worsen inflammation.

Age: It is most frequently diagnosed between the ages of 15 and 30




SYMPTOMS OF
ULCERATIVE COLITIS
Bloody Stool
Abdominal pain
Mucus or pus in stool
Nocturnal Defecations
Diarrhea
An urgent feeling to have a bowel movement
Fatigue
A constant urge to have a bowel movement even when the bowels are empty
Bay Clinic Of Chiropractic
Dr. Tony Salamay DC MS FASA
Functional Neurology & Clinical Nutrition
START LIVING AGAIN!


Severe
Surgery
Cyclosporine
Tacolimus
Infliximab
Moderate
AZA/6-MP
Corticosteroids
(topical/systemic)
Mild
Aminosalicylates
(topical/systemic)

Paralysis of Defecation in Persons With Spinal Cord Injuries

Paralysis of defecation, Paralysis of defecation refers to the loss of voluntary bowel control, usually due to nerve damage, spinal cord injuries, or neurological diseases. It leads to severe complications like chronic constipation, fecal impaction, or fecal incontinence.

known clinically as neurogenic bowel dysfunction (NBD), results from disrupted nerve signals between the brain and the gut.

It causes chronic constipation, fecal incontinence, or both.

Managing this requires a structured, personalized bowel program using scheduled timing, dietary adjustments, and mechanical or pharmacological stimulation to prevent impaction

When the spinal cord is injured somewhere between the conus medullaris and the brain, the voluntary portion of the defecation act is blocked while the basic cord reflex for defecation is still intact.

Nevertheless, loss of the voluntary aid to defecation—that is, loss of the increased abdominal pressure and relaxation of the voluntary anal sphincter—often makes defecation a difficult process in the person with this type of upper cord injury.

However, because the cord defecation reflex can still occur, a small enema to excite action of this cord reflex, usually given in the morning shortly after a meal, can often cause adequate defecation. In this way, people with spinal cord injuries that do not destroy the conus medullaris of the spinal cord can usually control their bowel movements each day.

General Disorders of the Gastrointestinal Tract

Vomiting

Vomiting is the forceful expulsion of stomach contents. It is usually a temporary defense mechanism triggered by common issues like food poisoning, stomach flu (gastroenteritis), motion sickness, or pregnancy.

The sensory signals that initiate vomiting originate mainly from the pharynx, esophagus, stomach, and upper portions of the small intestines.

As the nerve impulses are transmitted by vagal and sympathetic afferent nerve fibers to multiple distributed nuclei in the brain stem, especially the area postrema, that all together are called the "vomiting center."

At the onset of vomiting, strong intrinsic contractions occur in both the duodenum and the stomach, along with partial relaxation of the esophageal- stomach sphincter, thus allowing vomitus to begin moving from the stomach into the esophagus. From here, a specific vomiting act involving the abdominal muscles takes over and expels the vomitus to the exterior, as explained be low. Vomiting Act.

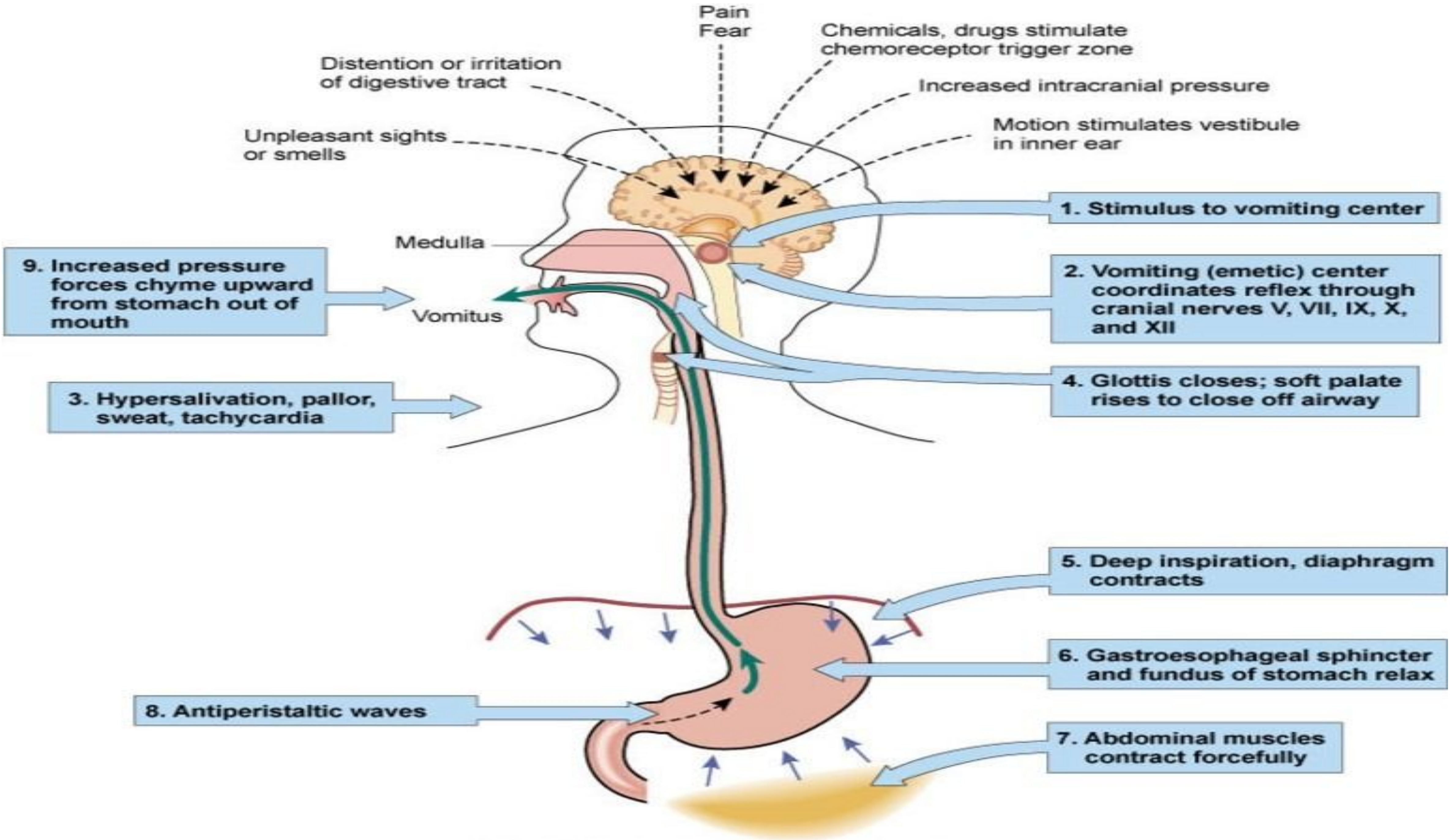
Once the vomiting center has been sufficiently stimulated and the vomiting act has been instituted, the first effects are the following: (1) a deep breath,

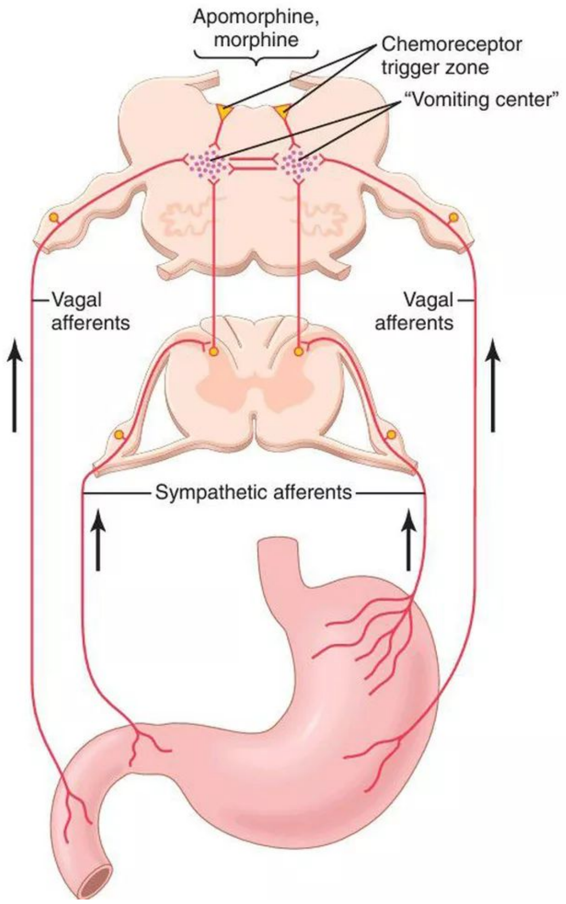
(2) raising of the hyoid bone and larynx to pull the upper esophageal sphincter open,

(3) closing of the glottis to pre vent vomitus flow into the lungs, and

(4) lifting of the soft palate to close the posterior nares.

Next comes a strong downward contraction of the diaphragm along with simultaneous contraction of all the abdominal wall muscles, which squeezes the stomach between the diaphragm and the abdominal muscles, building the intra gastric pressure to a high level. Finally, the lower esophageal sphincter relaxes completely, allowing expulsion of the gastric contents upward through the esophagus. Thus, the vomiting act results from a squeezing action of the muscles of the abdomen associated with simultaneous ous contraction of the stomach wall and opening of the esophageal sphincters so that the gastric contents can be expelled.


Pain
Fear
Chemicals, drugs stimulate chemoreceptor trigger zone
Distention or irritation of digestive tract
Increased intracranial pressure
Unpleasant sights or smells
Motion stimulates vestibule in inner ear
1. Stimulus to vomiting center
Medulla
9. Increased pressure forces chyme upward from stomach out of mouth
Vomitus
2. Vomiting (emetic) center coordinates reflex through cranial nerves V, VII, IX, X, and XII
4. Glottis closes; soft palate rises to close off airway
3. Hypersalivation, pallor, sweat, tachycardia
5. Deep inspiration, diaphragm contracts
6. Gastroesophageal sphincter and fundus of stomach relax
8. Antiperistaltic waves
7. Abdominal muscles contract forcefully



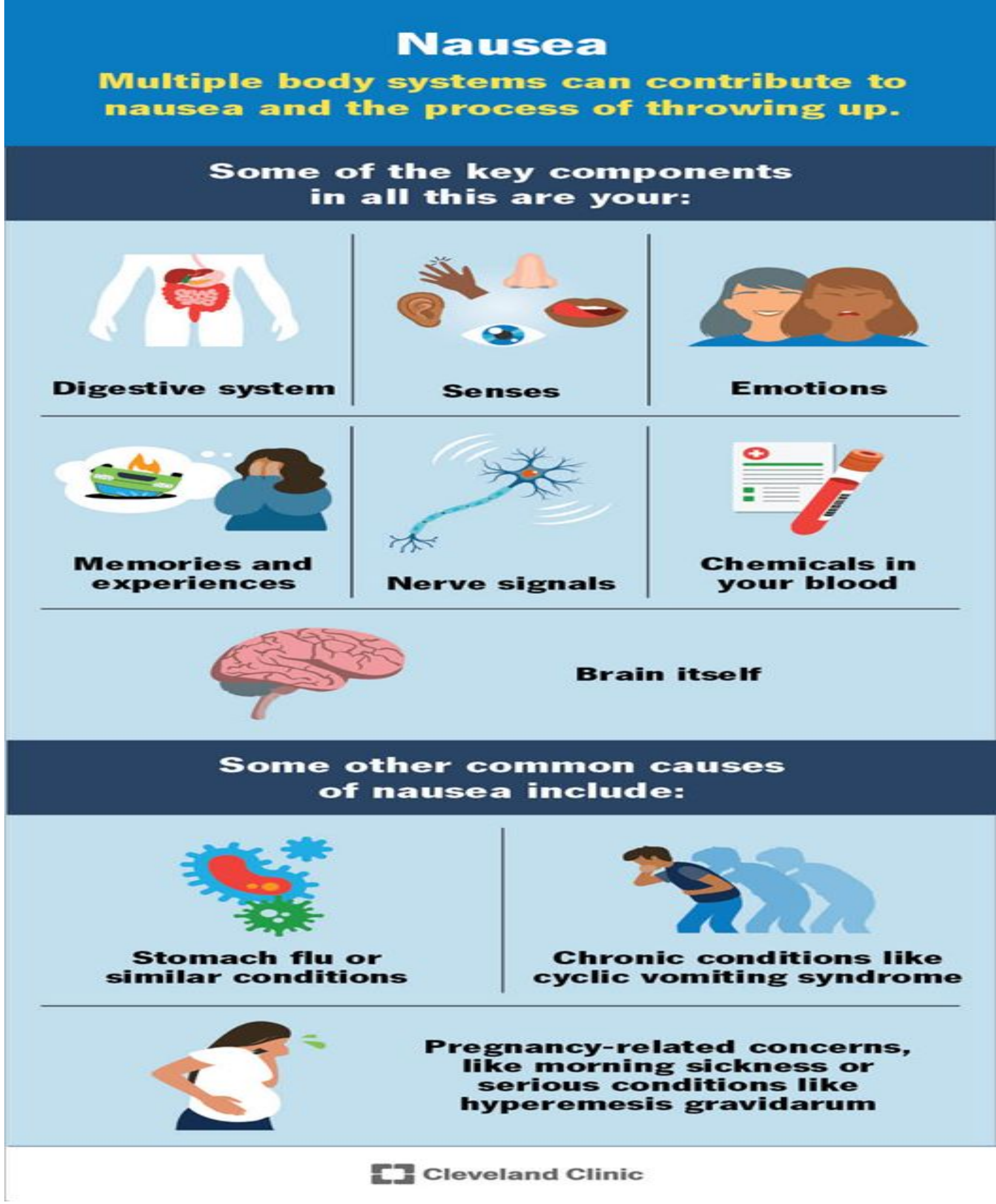
Apomorphine, morphine
Chemoreceptor trigger zone
"Vomiting center"
Vagal afferents
Vagal afferents
Sympathetic afferents

Nausea is an uneasy feeling in your stomach or chest that brings on the urge to vomit. It is a very common, non-specific symptom caused by a wide range of factors, including infections (stomach flu), motion sickness, migraines, and early pregnancy.

It can be caused by (1) irritative impulses coming from the gastrointestinal tract,

(2) impulses that originate in the lower brain associated with motion sickness, or

(3) impulses from the cerebral cortex to initiate vomiting. Vomiting occasionally occurs without the prodromal sensation of nausea, which indicates that only certain portions of the vomiting center are associated with the sensation of nausea



Gastrointestinal Obstruction

The gastrointestinal tract can become obstructed at almost any point along its course,

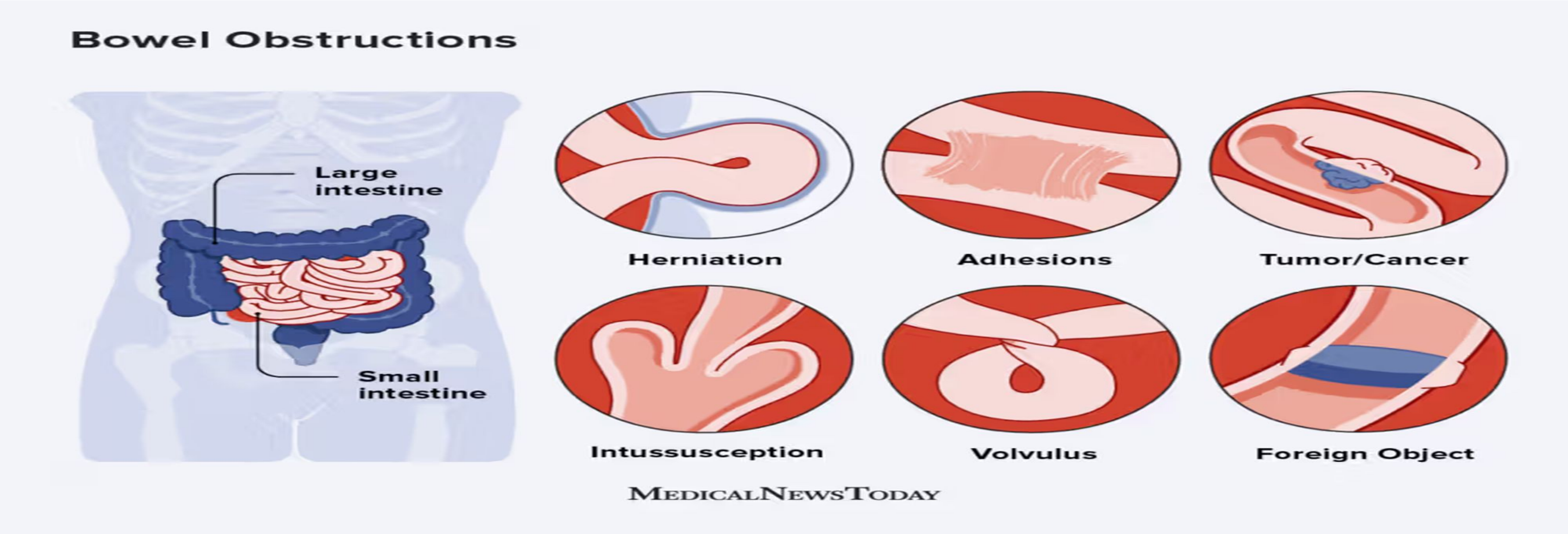
Some common causes of obstruction are the following:

(1) cancer,

(2) fibrotic constriction resulting from ulceration or from peritoneal adhesions,

(3) spasm of a gut segment, and

(4) paralysis of a gut segment.



The abnormal consequences of obstruction depend on the point in the gastrointestinal tract that becomes obstructed.

- **If the obstruction occurs at the pylorus**, which often results from fibrotic constriction after peptic ulceration, persistent vomiting of stomach contents occurs. This vomiting depresses bodily nutrition; it also causes excessive loss of hydrogen ions from the stomach and can result in various degrees of whole- body metabolic alkalosis.
- **If the obstruction is beyond the stomach**, antiperistaltic reflux from the small intestine causes intestinal juices to flow backward into the stomach, and these juices are vomited along with the stomach secretions. In this case, the person loses large amounts of water and electrolytes. He or she becomes severely dehydrated, but the loss of acid from the stomach and base from the small intestine may be approximately equal, so little change in acid–base balance occurs
- **If the obstruction is near the distal end of the large intestine**, feces can accumulate in the colon for a week or more.

The patient develops an intense feeling of constipation, but at first vomiting is not severe. After the large intestine has become completely filled and it finally becomes impossible for additional chyme to move from the small intestine into the large intestine, severe vomiting then occurs .

Prolonged obstruction of the large intestine can finally cause rupture of the intestine or dehydration and circulatory shock resulting from the severe vomiting.

Gases in the Gastrointestinal Tract (Flatus) Gases, called flatus, can enter the gastrointestinal tract from three sources:

1-swallowed air,

2-gases formed in the gut as a result of bacterial action, or

3-gases that diffuse from the blood into the gastrointestinal tract.

Most gases in the stomach are mixtures of nitrogen and oxygen derived from swallowed air. These gases are usually expelled by belching. Only small amounts of gas normally occur in the small intestine, and much of this gas is air that passes from the stomach into the intestinal tract.

In the large intestine, bacterial action generates most of the gases, including especially carbon dioxide, methane, and hydrogen. When methane and hydrogen become suitably mixed with oxygen, an actual explosive mixture is sometimes formed. Use of the electric cautery during sigmoidoscopy has been known to cause a mild explosion.

Certain foods are known to cause greater expulsion of flatus through the anus than others—beans, cabbage, on ion, cauliflower, corn, and certain irritant foods such as vinegar.

Some of these foods serve as a suitable medium for gas-forming bacteria, especially unabsorbed ferment able types of carbohydrates.

For example, beans contain an indigestible carbohydrate that passes into the colon and becomes a superior food for colonic bacteria. But in other cases, excess expulsion of gas results from irritation of the large intestine, which promotes rapid peristaltic expulsion of gases through the anus before they can be absorbed.

The amount of gases entering or forming in the large intestine each day averages 7 to 10 liters, whereas the average amount expelled through the anus is usually only about 0.6 liter. The remainder is normally absorbed into the blood through the intestinal mucosa and expelled through the lungs

Thank you